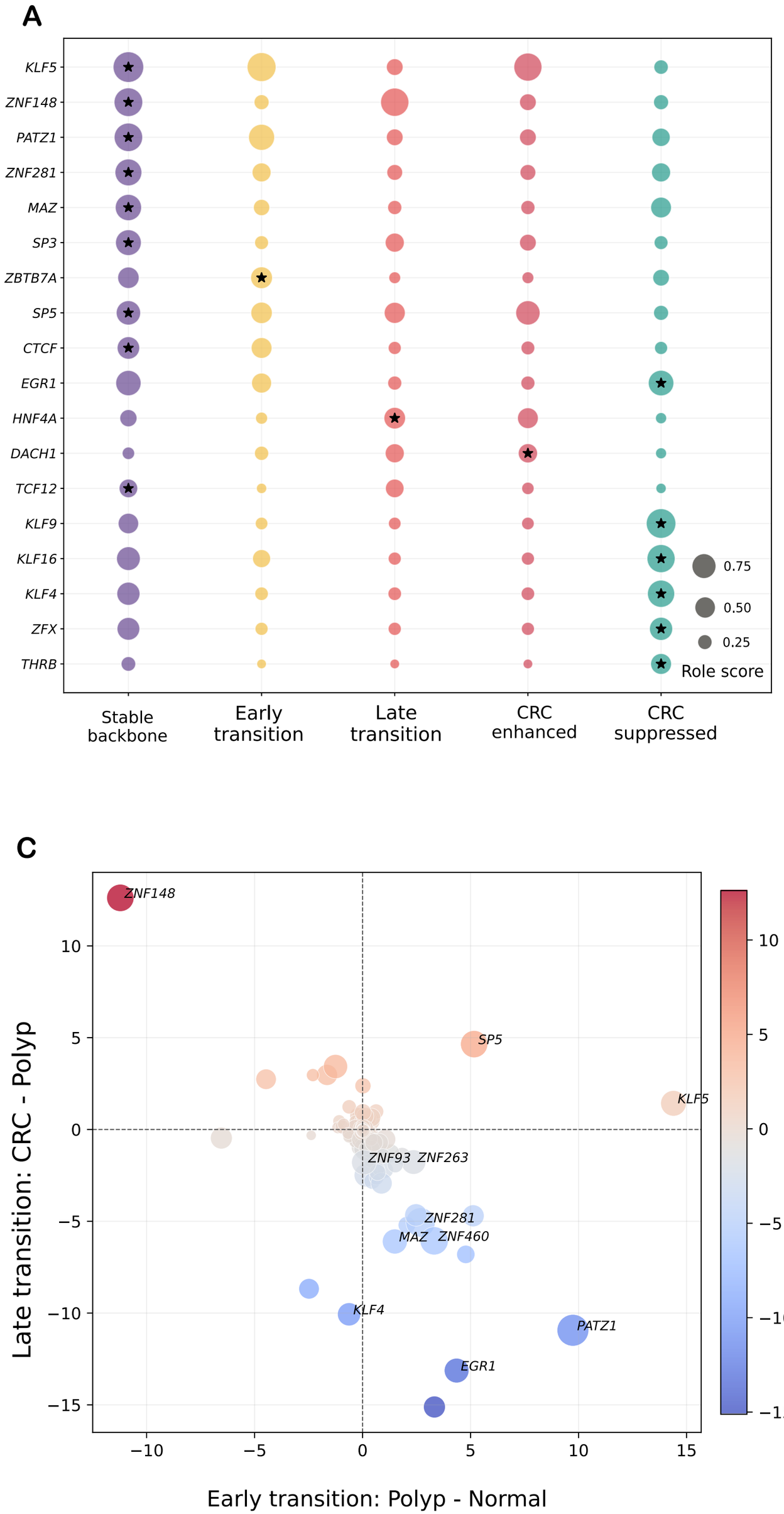
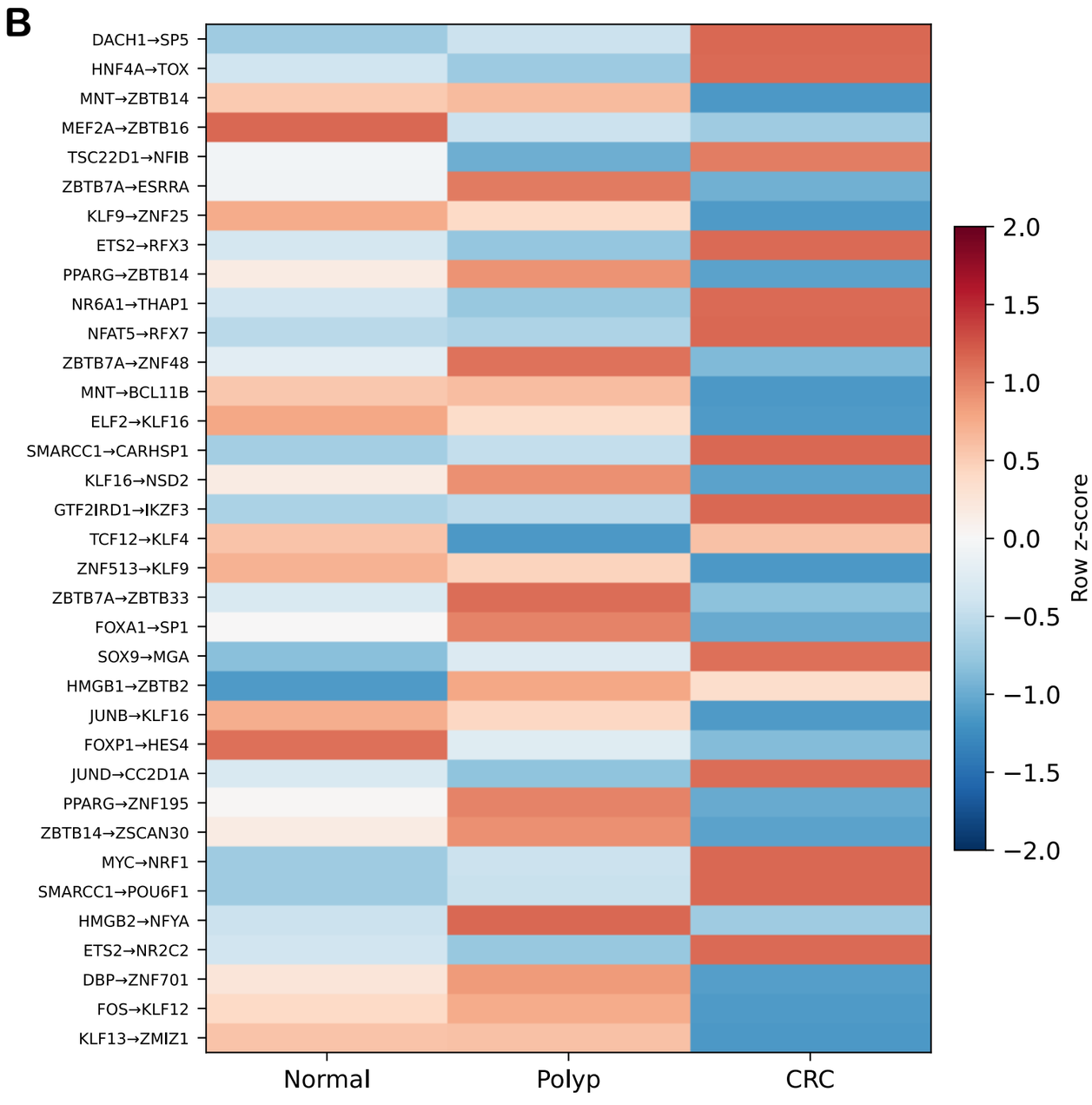
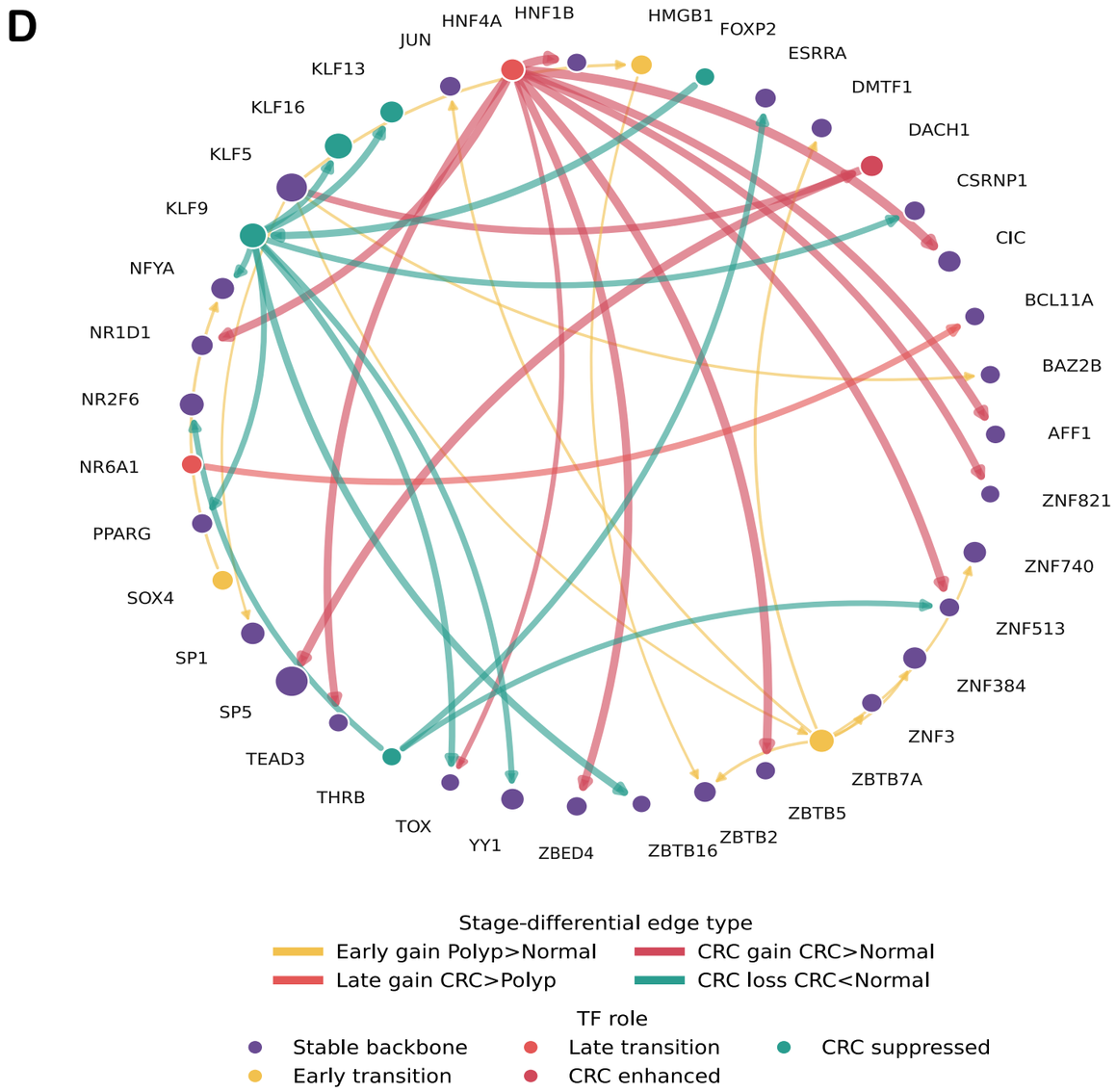

A
KLF5
ZNF148
PATZ1
ZNF281
MAZ
SP3
ZBTB7A
SP5
CTCF
EGR1
HNF4A
DACH1
TCF12
KLF9
KLF16
KLF4
ZFX
THRB
0.75
0.50
0.25
Role score
Stable backbone
Early transition
Late transition
CRC enhanced
CRC suppressed


B
DACH1→SP5
HNF4A→TOX
MNT→ZBTB14
MEF2A→ZBTB16
TSC22D1→NFIB
ZBTB7A→ESRRA
KLF9→ZNF25
ETS2→RFX3
PPARG→ZBTB14
NR6A1→THAP1
NFAT5→RFX7
ZBTB7A→ZNF48
MNT→BCL11B
ELF2→KLF16
SMARCC1→CARHSP1
KLF16→NSD2
GTF2IRD1→IKZF3
TCF12→KLF4
ZNF513→KLF9
ZBTB7A→ZBTB33
FOXA1→SP1
SOX9→MGA
HMGB1→ZBTB2
JUNB→KLF16
FOXP1→HES4
JUND→CC2D1A
PPARG→ZNF195
ZBTB14→ZSCAN30
MYC→NRF1
SMARCC1→POU6F1
HMGB2→NFYA
ETS2→NR2C2
DBP→ZNF701
FOS→KLF12
KLF13→ZMIZ1
Normal
Polyp
CRC
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Row z-score

C
ZNF148
SP5
KLF5
ZNF93
ZNF263
ZNF281
MAZ
ZNF460
KLF4
PATZ1
EGR1
10
5
0
−5
−10
−15
−10
−5
0
5
10
15
Late transition: CRC - Polyp
Early transition: Polyp - Normal
Late transition delta


D
HNF4A
HNF1B
HMGB1
FOXP2
JUN
ESRRA
KLF13
DMTF1
KLF16
DACH1
KLF5
CSRNP1
KLF9
CIC
NFYA
BCL11A
NR1D1
BAZ2B
NR2F6
AFF1
NR6A1
ZNF821
PPARG
ZNF740
SOX4
ZNF513
SP1
ZNF384
SP5
ZNF3
TEAD3
ZBTB7A
THRB
ZBTB5
TOX
ZBTB2
YY1
ZBED4
ZBTB16
Stage-differential edge type
Early gain Polyp>Normal
CRC gain CRC>Normal
Late gain CRC>Polyp
CRC loss CRC<Normal
TF role
Stable backbone
Late transition
CRC suppressed
Early transition
CRC enhanced